

Rapport de mobilité Erasmus+ (SERE)

Semestre 8 - Mobilité d'études

Ilyas BOUNOUA

École Centrale de Nantes - UCLouvain (EPL)

Période: 01/02/2025 – 30/06/2025



Campus de l'Université catholique de Louvain (Louvain-la-Neuve)

École Centrale de Nantes

Direction des Relations Internationales

Fiche établissement d'accueil

Année universitaire :

Elève ingénieur Nom: Bounoua Prénom : Ilyas

Période du séjour à l'étranger de : 01-02-2025 à : 30-06-2025

Type de séjour à l'étranger ⁽¹⁾:

☐	Stage de Connaissance du Monde de l'Entreprise	
☐	Stage ingénieur	
☐	Travail de fin d'études	Option :
☒	S8	☐ SE ☐ Césure
☐	DD	

Renseignements sur l'établissement d'accueil :

Type l'établissement ⁽¹⁾ :	☐ Entreprise	☒ université	☐ autre :
Nom de l'établissement : Université catholique de Louvain			
Adresse		Ville	Pays
Pl. de l'Université 1, 1348		Ottignies-Louvain -la-Neuve	Belgique
Tél	+3210472111	Fax	
site web	https://www.uclouvain.be/fr		
Département d'accueil : EPL - Ecole Polytechnique de Louvain			
Votre responsable directe (encadrant) : Charles Pecheur			
Tél	+32 10 47 87 79	Fax	+32 10 45 03 45
E.mail	Charles.Pecheur@uclouvain.be		
Autres informations utiles pour les élèves qui recherchent un stage ou un TFE :			

Renseignements sur le travail réalisé pendant le séjour (stage, TFE, thèse master):

Titre du stage ou TFE :
Résumé en 2 lignes :
Degré de satisfaction de la mobilité et de l'encadrement dans l'établissement d'accueil (0 à 5)

Autres commentaires :

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¹ Cocher la case correspondante

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I Présentation générale du pays et de la ville d'accueil

I.1 Contexte géographique et linguistique

La Belgique est un pays multilingue et multiculturel. Louvain-la-Neuve se situe en Wallonie (région francophone) et la langue dominante dans la vie quotidienne et universitaire est le français. Le campus de Louvain-la-Neuve est une ville universitaire planifiée, piétonne en grande partie, qui offre un cadre de vie sûr et des infrastructures pour les étudiants.

I.2 Culture et vie quotidienne

Louvain-la-Neuve est une ville étudiante : bars, restaurants, bibliothèques universitaires et activités culturelles sont très présents. La proximité de Bruxelles facilite l'accès à un large choix de transports et d'animations.

I.3 Déplacements

Transports régionaux : trains (SNCB) et réseau de bus régional (TEC) desservent Louvain-la-Neuve. Les aéroports les plus utilisés pour nous les échanges Erasmus sont Bruxelles-National, Charleroi (via train+bus), et Bruxelles South Charleroi (navettes), accessibles en 30–90 minutes selon l'itinéraire.



Place universitaire de Louvain-la-Neuve

II Procédures administratives et installation

II.1 Avant le départ

Avant le départ, j'ai complété le Learning Agreement officiel (Erasmus) avec les cours choisis et obtenu les signatures nécessaires. Le document "Learning Agreement" (version pré-mobilité et modifications) est joint dans les documents administratifs.

II.2 Arrivée et inscription à l'université

L'inscription à UCLouvain (EPL) s'est faite via la page d'accueil "Exchange application" et le bureau des relations internationales d'EPL. Il est important de respecter les dates d'ouverture d'inscription (généralement : semestre d'hiver 14 sept. - 31 janv. ; pour notre cas le semestre de printemps 1er fév. - 30 juin) et de préparer les pièces justificatives (passeport, assurance, Learning Agreement signé, etc.).

II.3 Logement

Des solutions : résidences universitaires, logements privés, ou colocations. La cellule logement d'UCLouvain propose des ressources et listes pour les étudiants entrants. Je recommande de s'inscrire sur les listes d'attente le plus tôt possible.

III Semestre S8 - Louvain School of Engineering (EPL)

III.1 Présentation de l'unité d'accueil

La Louvain School of Engineering (EPL) fait partie d'UCLouvain et accueille de nombreux programmes d'ingénierie. Le campus de Louvain-la-Neuve est un environnement de campus étendu, calme et majoritairement piéton. Les cours sont en anglais ("English-friendly" ou English).

III.2 Déroulé des semestres et évaluations

Le calendrier officiel indique les dates de semestre (période de mobilité indiquée à la page de l'EPL). Les modes d'évaluation varient par cours : projets informatiques, devoirs, partiels finaux, présentations orales, et projets en groupe.

III.3 Modules suivis et apports pédagogiques

Ci-dessous : liste des cours suivis durant le semestre et un court commentaire sur leur contenu et l'apport pour ma formation.

Code / Intitulé	Commentaire / Apport
LINFO2262 – Machine Learning : Classification and Evaluation	Cours axé sur les méthodes de classification, l'évaluation et les projets informatiques pratiques (compétences utiles pour la mise en place et l'évaluation d'algorithmes de ML).
LINFO2275 – Data mining & decision making	Approfondissement des techniques de fouille de données et leur application à la décision ; utile pour projets de décision basée sur les données.
LINFO2365 – Constraint Programming	Méthodes de résolution de contraintes (CSP), langages et solvers ; complète les méthodes d'optimisation et modélisation.
LINMA2710 – Scientific computing	Outils numériques, programmation scientifique (C++, Python, Julia), et techniques de calcul pour simulation et optimisation.
LINFO2364 – Mining Patterns In Data	Techniques avancées de recherche de motifs et pattern mining ; utile pour analyser grands jeux de données.
LINFO2335 – Programming paradigms	Vision historique et théorique des paradigmes de programmation (fonctionnel, orienté objet, déclaratif...) – utile pour mieux choisir les outils de développement.

Remarque : Les descriptifs de ces cours (objectifs, contenu, enseignant, langue) sont disponibles dans le catalogue de cours d'UCLouvain (pages des cours LINFO2262, LINFO2275, LINFO2365, LINMA2710, LINFO2364, LINFO2335).

III.4 Particularités de l'enseignement et apports pour le projet professionnel

L'approche pédagogique à UCLouvain combine la théorie et beaucoup de pratique : projets, travaux en groupe, et utilisations d'outils (Inginious, Moodle, projets informatiques réels). Les cours en anglais favorisent l'insertion dans des contextes internationaux et la collaboration. L'expérience renforce mes compétences en apprentissage automatique, en calcul scientifique et en fouille de données, directement alignées avec mon parcours d'ingénieur et mes perspectives professionnelles en data science / recherche.

IV Conclusion

La mobilité Erasmus+ à UCLouvain (EPL) m'a permis d'approfondir des compétences techniques (ML, data mining, calcul scientifique, programmation avancée) tout en me confrontant à un environnement international et à des méthodes pédagogiques différentes. Je recommande de profiter pleinement des ressources locales (bibliothèque, groupes d'étude, ateliers) et de documenter ses projets pour la reconnaissance académique au retour.

Annexes

- Learning Agreement (signature pré-mobilité et modifications) – document administratifs.
- Transfert des crédits et relevé de notes (Transcript of Records)

Learning Agreement Student Mobility for Studies

Higher Education:
Learning Agreement form
Academic Year 2024/2025

Student	Last name(s)	First name(s)	Date of birth	Nationality ¹	Sex [M/F]	Study cycle ²	Field of education ³
	BOUNOUA	Ilyas	10/01/2004	Marocaine	M	Master	07 – Engineering,
Sending Institution	Name	Faculty/Department	Erasmus code ⁴ (if applicable)	Address	Country	Contact person name ⁵ ; email; phone	
	Centrale NANTES	General engineer	F NANTES 07	1 Rue de la Noë, 44300 Nantes	France	Clementine Houssais Clementine.Houssais@ec-nantes.fr +33 2 40 37 15 80	
Receiving Institution	Name	Faculty/ Department	Erasmus code (if applicable)	Address	Country	Contact person name; email; phone	
	UCLouvain	Louvain School of Engineering (EPL)	B LOUVAIN01	Rue Archimède 1 bte L6.11.01 1348 Louvain-la-Neuve	Belgium	Sandrine Matthieu epl-exchange@uclouvain.be +32 10 47 85 30	

Before the mobility

Study Programme at the Receiving Institution				
Planned period of the mobility: from [month/year] __/202__ to [month/year] __/202__				
Table A Before the mobility	Component ⁶ code (if any)	Component title at the Receiving Institution (as indicated in the course catalogue ⁷)	Semester [e.g. autumn/spring; term]	Number of ECTS credits (or equivalent) ⁸ to be awarded by the Receiving Institution upon successful completion
	LINFO2262	Machine Learning: Classification and Evaluation	Spring	6
	LINFO2275	Data mining & decision making	Spring	5
	LINFO2365	Constraint Programming	Spring	5
	LINMA2710	Scientific computing	Spring	5
	LINFO2144	Secured systems engineering	Spring	5
	LINFO2335	Programming paradigms	Spring	5
			Spring	
				Total: 31
Web link to the course catalogue at the Receiving Institution describing the learning outcomes: [web link to the relevant information]				

The level of language competence⁹ in English that the student already has or agrees to acquire by the start of the study period is C1.

Recognition at the Sending Institution				
Table B Before the mobility	Component code (if any)	Component title at the Sending Institution (as indicated in the course catalogue)	Semester [e.g. autumn/spring; term]	Number of ECTS credits (or equivalent) to be recognised by the Sending Institution
	ARSIG	analysis and representation of signals	Autumn	3
	ATRIM	image analysis and processing	Autumn	3
	CSOPT	scientific computation and optimization	Autumn	3
	MSTAT	statistical modeling and learning	Autumn	3
	GRASP	graphs and time series analysis	Autumn	3
	APSTA	statistical learning	Autumn	3
	IMINV	imaging and inverse methods	Autumn	3
	FEBIO	filtering and processing of biomedical signals	Autumn	3
	PRTSI	project in signal and image processing	Autumn	3
				Total:
Provisions applying if the student does not complete successfully some educational components: [web link to the relevant information]				

Learning Agreement Student Mobility for Studies

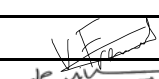
Higher Education:
Learning Agreement form
Academic Year 2024/2025

Commitment					
By signing this document, the student, the Sending Institution and the Receiving Institution confirm that they approve the Learning Agreement and that they will comply with all the arrangements agreed by all parties. Sending and Receiving Institutions undertake to apply all the principles of the Erasmus Charter for Higher Education relating to mobility for studies (or the principles agreed in the Inter-Institutional Agreement for institutions located in Partner Countries). The Beneficiary Institution and the student should also commit to what is set out in the Erasmus+ grant agreement. The Receiving Institution confirms that the educational components listed in Table A are in line with its course catalogue and should be available to the student. The Sending Institution commits to recognise all the credits or equivalent units gained at the Receiving Institution for the successfully completed educational components and to count them towards the student's degree as described in Table B. Any exceptions to this rule are documented in an annex of this Learning Agreement and agreed by all parties. The student and the Receiving Institution will communicate to the Sending Institution any problems or changes regarding the study programme, responsible persons and/or study period.					
Commitment	Name	Email	Position	Date	Signature
Student	Ilyas BOUNOUA	Ilyasbounoua9@outlook.fr	Student	20/10/2024	
Responsible person ¹⁰ at the Sending Institution	Clémentine Houssais Aurelien SERANDOUR	CLEMENTINE.HOUSSAIS@EC-NANTES.FR	International mobility manager	24/10/2024	
Responsible person ¹¹ at the Receiving Institution					

During the Mobility

Exceptional changes to Table A (to be approved by e-mail or signature by the student, the responsible person in the Sending Institution and the responsible person in the Receiving Institution)							
Table A2 During the mobility	Component code (if any)	Component title at the Receiving Institution (as indicated in the course catalogue)	Kept component [tick if applicable]	Deleted component [tick if applicable]	Added component [tick if applicable]	Reason for change ¹²	Number of ECTS credits (or equivalent)
	LINFO2262	Machine Learning: Classification and Evaluation	X	☐	☐		6
	LINFO2275	Data mining & decision making	X	☐	☐		5
	LINFO2365	Constraint Programming	X	☐	☐		5
	LINMA2710	Scientific computing	X	☐	☐		5
	LINFO2144	Secured systems engineering	☐	X	☐		5
	LINFO2364	Mining Patterns In Data	☐	☐	X	5	5
	LINFO2335	Programming paradigms	X	☐	☐		5
							Total: 31

Exceptional changes to Table B (if applicable) (to be approved by e-mail or signature by the student and the responsible person in the Sending Institution)						
Table B2 During the mobility	Component code (if any)	Component title at the Sending Institution (as indicated in the course catalogue)	Kept component [tick if applicable]	Deleted component [tick if applicable]	Added component [tick if applicable]	Number of ECTS credits (or equivalent)
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	
			☐	☐	☐	

Commitment	Name	Email	Position	Date	Signature
Student			Student		
Responsible person ¹⁰ at the Sending Institution	Vincent FREMONT	international@ec-nantes.fr	Dean of International Relations	14/02/2025	
Responsible person ¹¹ at the Receiving Institution	Charles Pecheur	charles.pecheur@uclouvain.be	Head of INFO programme	13/2/25	

After the Mobility

Transcript of Records at the Receiving Institution					
Start and end dates of the study period: from [day/month/year] to [day/month/year]					
Table C After the mobility	Component code (if any)	Component title at the Receiving Institution (as indicated in the course catalogue)	Was the component successfully completed by the student? [Yes/No]	Number of ECTS credits (or equivalent)	Grades received at the Receiving Institution
				Total: ...	

Transcript of Records and Recognition at the Sending Institution				
Start and end dates of the study period: from [day/month/year] to [day/month/year]				
Table D After the mobility	Component code (if any)	Title of recognised component at the Sending Institution (as indicated in the course catalogue)	Number of ECTS credits (or equivalent) recognised	Grades registered at the Sending Institution (if applicable)
			Total: ...	

¹ **Nationality:** country to which the person belongs administratively and that issues the ID card and/or passport.

² **Study cycle:** Short cycle (EQF level 5) / Bachelor or equivalent first cycle (EQF level 6) / Master or equivalent second cycle (EQF level 7) / Doctorate or equivalent third cycle (EQF level 8).

³ **Field of education:** The [ISCED-F 2013 search tool](http://ec.europa.eu/education/tools/iscsed-f_en.htm) available at http://ec.europa.eu/education/tools/iscsed-f_en.htm should be used to find the ISCED 2013 detailed field of education and training that is closest to the subject of the degree to be awarded to the student by the Sending Institution.

⁴ **Erasmus code:** a unique identifier that every higher education institution that has been awarded with the Erasmus Charter for Higher Education (ECHE) receives. It is only applicable to higher education institutions located in Programme Countries.

⁵ **Contact person:** person who provides a link for administrative information and who, depending on the structure of the higher education institution, may be the departmental coordinator or works at the international relations office or equivalent body within the institution.

⁶ An "educational component" is a self-contained and formal structured learning experience that features learning outcomes, credits and forms of assessment. Examples of educational components are: a course, module, seminar, laboratory work, practical work, preparation/research for a thesis, mobility window or free electives.

⁷ **Course catalogue:** detailed, user-friendly and up-to-date information on the institution's learning environment that should be available to students before the mobility period and throughout their studies to enable them to make the right choices and use their time most efficiently. The information concerns, for example, the qualifications offered, the learning, teaching and assessment procedures, the level of programmes, the individual educational components and the learning resources. The Course Catalogue should include the names of people to contact, with information about how, when and where to contact them.

⁸ **ECTS credits (or equivalent):** in countries where the "ECTS" system is not in place, in particular for institutions located in Partner Countries not participating in the Bologna process, "ECTS" needs to be replaced in the relevant tables by the name of the equivalent system that is used, and a web link to an explanation to the system should be added.

⁹ **Level of language competence:** a description of the European Language Levels (CEFR) is available at: <https://europass.cedefop.europa.eu/en/resources/european-language-levels-cefr>

¹⁰ **Responsible person at the Sending Institution:** an academic who has the authority to approve the Learning Agreement, to exceptionally amend it when it is needed, as well as to guarantee full recognition of such programme on behalf of the responsible academic body. The name and email of the Responsible person must be filled in only in case it differs from that of the Contact person mentioned at the top of the document.

¹¹ **Responsible person at the Receiving Institution:** the name and email of the Responsible person must be filled in only in case it differs from that of the Contact person mentioned at the top of the document.

¹² **Reasons for exceptional changes to study programme abroad (choose an item number from the table below):**

Reasons for deleting a component	Reason for adding a component
1. Previously selected educational component is not available at the Receiving	5. Substituting a deleted
2. Component is in a different language than previously specified in the course	6. Extending the mobility period
3. Timetable conflict	7. Other (please specify)
4. Other (please specify)	

RELEVÉ DE NOTES - TRANSCRIPT OF RECORDS

ECTS - EUROPEAN CREDIT TRANSFER SYSTEM

NAME OF INSTITUTION: Université catholique de Louvain - B LOUVAIN01 Faculty/Department: Ecole polytechnique de Louvain Rue Archimède, 1/L6.11.01 1348 Louvain-la-Neuve Tél: 010472460	
NAME OF SENDING INSTITUTION: Ecole Centrale de Nantes France	
NAME OF STUDENT: Bounoua First name: Ilyas Date and place of birth: 10 January 2004 at RABAT (sex): M Matriculation date: 25 November 2024 Matriculation number: 28372410	

Course Unit code (1)	Title of the course unit	Duration of course unit (2)	Language of instruction	Local grade (3)	ECTS grade (4)	ECTS credits (5)
Academic year 2024-2025: <i>Incoming mobility EPL</i>						
LINFO2365	Constraint programming (Constraint programming) [30h - 15h]		ENG	15/20	B	5
LINFO2275	Data mining and decision making (Data mining & decision making) [30h - 15h]		ENG	11/20	D	5
LINFO2262	Machine Learning : classification and evaluation (Machine Learning :classification and evaluation) [30h - 30h]		ENG	14/20	C	6
LINFO2364	Mining Patterns in Data (Mining Patterns in Data) [30h - 15h]		ENG	10/20	E	5
LINFO2335	Programming paradigms (Programming paradigms) [30h - 15h]		ENG	14/20	C	5
LINMA2710	Scientific computing (Scientific computing) [30h - 22.5h]		ENG	15/20	B	5

(1) (2) (3) (4) (5) conferatur infra

Date

09 July 2025



(1) Course unit code:

Refer to the Course research via
<http://www.uclouvain.be>

(2) Duration of course unit:

Y = full academic year S = semester

(3) Description of the institutional grading system:

The final assessment of a course will result in a mark on a scale from 0 to 20, the minimum pass mark being 10/20.
 Marks followed by V or W are not included in the average mark.
 NE = Not evaluated
 T = Cheating
 Abs. = Absence

(4) ECTS grading table:

UCLouvain uses an ECTS grading table based on the total number of successful grades obtained in all courses of the same study cycle in the same faculty over the last five academic years. It is reviewed each year.

Grade	20	19	18	17	16	15	14	13	12	11	10	< 10
% of students awarded the grade	2.6	2.9	6.7	10.1	15.3	14.8	13.9	11.3	9.0	6.9	6.5	-
ECTS grade	A				B		C		D		E	F

(5) ECTS credits:

1 full academic year = 60 credits
 1 semester = 30 credits